

# Project ABEONA Crew Capsule Main Engine: Thrust Chamber, Feed System, and Pressurization Design

[Your Name]

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## 1 Introduction

This report presents the preliminary design of the single 25 kN main engine of the Project ABEONA Crew Capsule service module (SM), together with its propellant feed and helium pressurization system and the Onshape CAD model of the propulsion bay. The engine burns nitrogen tetroxide (NTO) and monomethylhydrazine (MMH), is pressure-fed by regulated helium and operates only in vacuum. Vehicle requirements come from the Crew Capsule tab of the mission breakdown spreadsheet [5]; the propellant choice and the SM envelope come from the Preliminary Design Review (PDR) [5,6]. All results are analytical predictions from the design scripts; none has yet been demonstrated by test.

## 2 Requirements

Table 1: Crew Capsule requirements (mission breakdown spreadsheet [5]).

| Parameter                                            | Value                                                              |
|------------------------------------------------------|--------------------------------------------------------------------|
| Initial / propellant / structural / payload mass, kg | 38,000 / 21,000 / 6,500 / 10,500                                   |
| Burnout mass, kg                                     | 17,000                                                             |
| Minimum $I_{sp}$ , s                                 | 280                                                                |
| $\Delta v$ , m/s                                     | 2,209                                                              |
| Thrust; number of engines                            | 25 kN; 1                                                           |
| Mass flow and burn time at 280 s                     | 9.10 kg/s; 2,307 s                                                 |
| $T/W$ at the Moon, initial / burnout                 | 0.405 / 0.905                                                      |
| Propellant densities (MMH / NTO), kg/m <sup>3</sup>  | 875 / 1,440                                                        |
| SM envelope (PDR)                                    | 4.98 m diameter $\times$ 6 m; nozzle exit 2.01 m below the SM base |

### 3 Design Baseline

Table 2: Design baseline.

| Item               | Selection                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| Propellants        | NTO / MMH, O/F 1.65 (PDR; gives equal tank volumes with the specified densities)                                     |
| Cycle              | Pressure-fed, regulated helium (AJ10-190 / European Service Module heritage; no turbomachinery, simple restart)      |
| Chamber pressure   | 0.862 MPa nozzle-entrance stagnation pressure (AJ10-190 value, where the efficiency calibration applies)             |
| Nozzle             | 80 % Rao optimum-thrust bell, area ratio 108, set by the 2.642 m engine-length envelope                              |
| Thermal protection | Silica-phenolic ablative chamber and throat; fuel-rich outer injector ring; radiation-cooled C-103 extension         |
| Injector           | Unlike-doublet impinging elements with 12 quarter-wave acoustic cavities                                             |
| Tanks              | One Ti-6Al-4V sphere per propellant, NTO below MMH, screen-channel propellant management device (PMD) in each        |
| Pressurization     | Helium at 31 MPa in four composite overwrapped pressure vessels (COPVs); two series regulators; tank ullage 1.28 MPa |

Principal assumptions: ideal-gas combustion products in chemical equilibrium;  $g = 9.81 \text{ m/s}^2$  (as in the spreadsheet); propellants at 293 K nominal (283–305 K range); delivered efficiency calibrated to the AJ10-190 flight value (nominal) and the R-4D-11 (pessimistic) [9,10]; gas-side heat transfer by Bartz [3] with a 1.3 factor for sizing; ablator properties, material allowables, component pressure drops and component masses as stated, to be verified. Combustion performance was computed with an equilibrium solver implementing the NASA CEA method [2] with the CEA thermodynamic database [11], with liquid MMH and NTO properties from RocketProps [8]; it reproduces published CEA results within 0.02 % and is cross-checked against RocketCEA in Sec. 4.3.

## 4 Performance

### 4.1 Ideal and delivered performance

Table 3: Ideal vacuum performance at 0.862 MPa, O/F 1.65, area ratio 108.

| Quantity          | Shifting equilibrium | Frozen at chamber |
|-------------------|----------------------|-------------------|
| $T_c$ , K         | 3,042                | 3,042             |
| $c^*$ , m/s       | 1,737.9              | 1,698.6           |
| $C_{F,vac}$       | 1.928                | 1.875             |
| $I_{sp,vac}$ , s  | 341.5                | 324.7             |
| Exit pressure, Pa | 430                  | 331               |

The delivered specific impulse is built up from the ideal shifting-equilibrium value (Table 4). The efficiency  $\eta_{I_{sp}}$  is the ratio of flight to ideal  $I_{sp}$  for the reference engine at its own conditions, and includes combustion, divergence, boundary-layer and kinetic losses. The fuel-rich outer ring (Sec. 6) is a separate penalty.

$$I_{sp,d} = I_{sp,ideal} \eta_{I_{sp}} \eta_{BZ}, \quad c_d^* = c_{ideal}^* \eta_{c^*} \eta_{c^*,BZ} \quad (1)$$

Table 4: Delivered performance build-up.

| Step                                                   | Nominal                                       |                                 | Pessimistic  |
|--------------------------------------------------------|-----------------------------------------------|---------------------------------|--------------|
| Ideal $I_{sp,vac}$ (shifting, $\varepsilon = 108$ ), s | 341.5                                         |                                 | 341.5        |
| $\eta_{I_{sp}}$ (reference engine)                     | 0.9417 (AJ10-190: 315.1 / 334.6 s)            | 0.8908 (R-4D-11: 310 / 348.0 s) |              |
| After efficiency, s                                    | 321.6                                         |                                 | 304.2        |
| Fuel-rich ring, $\eta_{BZ}$                            | 0.9976 ( $-0.78$ s)                           |                                 | 0.9976       |
| <b>Delivered <math>I_{sp,vac}</math>, s</b>            | <b>320.8</b>                                  |                                 | <b>303.5</b> |
| Delivered $c^*$ , m/s                                  | $1,737.9 \times 0.99 \times 0.9736 = 1,675.1$ |                                 | –            |
| Delivered $C_{F,vac}$                                  | 1.879                                         |                                 | –            |
| Mass flow at 25 kN, kg/s                               | 7.943 (NTO 4.946, MMH 2.998)                  |                                 | 8.397        |
| Firing time for 21,000 kg, s                           | 2,644                                         |                                 | 2,501        |

## 4.2 Mission firing profile

The 2,644 s is the cumulative firing time to expend the full 21,000 kg load, not one continuous burn. The polar mission requires 9 burns (near-rectilinear halo orbit insertion and departure, two orbit transfers, three rendezvous burns, two trajectory corrections) plus 2 abort restarts, 11 starts in total. The engine is designed for 22 starts and would be qualified for 44. The longest single burn is taken as 1,250 s for the injector soak-back check.

## 4.3 Verification with RocketCEA

The ideal design point was re-run with RocketCEA 1.2.3 [7] using the same heats of formation and 293.15 K propellant temperature (Table 5). Every quantity agrees within 0.3 %.

Table 5: Ideal performance: design solver against RocketCEA (same basis).

| Quantity                    | Design solver | RocketCEA | Difference |
|-----------------------------|---------------|-----------|------------|
| $T_c$ , K                   | 3,042.3       | 3,045.7   | +0.11 %    |
| Chamber molar mass, g/mol   | 20.42         | 20.41     | −0.03 %    |
| $c^*$ , shifting, m/s       | 1,737.9       | 1,739.4   | +0.08 %    |
| $I_{sp,vac}$ , shifting, s  | 341.50        | 341.99    | +0.14 %    |
| $I_{sp,vac}$ , frozen, s    | 324.71        | 325.04    | +0.10 %    |
| Exit pressure, shifting, Pa | 429.9         | 430.3     | +0.09 %    |

## 5 Thrust Chamber and Nozzle Geometry

The nozzle contour was generated with an axisymmetric method-of-characteristics solver [12] and optimized for vacuum thrust at fixed length and exit diameter (Rao’s method [4]). The solver reproduces the analytical divergence efficiency of a  $15^\circ$  cone (0.98397 against 0.98296) and a two-dimensional wedge (0.98882 against 0.98862) with 0.11 % mass-flow error. The exit pressure of 430 Pa (0.42 % of sea-level pressure) confirms a vacuum-only nozzle; the flow would separate at sea level. Shortening the bell to 70 % and enlarging the area ratio within the same engine length gains only 0.1 s, within the accuracy of the method, so the 80 % bell was kept.

Table 6: Thrust-chamber and nozzle geometry.

| Parameter                                                   | Value                            | Basis                     |
|-------------------------------------------------------------|----------------------------------|---------------------------|
| Throat area / diameter                                      | 154.4 cm <sup>2</sup> / 140.2 mm | $\dot{m}c_d^*/P_c$        |
| Exit diameter / area ratio                                  | 1.457 m / 108.0                  | $(D_e/D_t)^2$             |
| Contraction ratio / chamber diameter                        | 3.0 / 242.8 mm                   | assumed                   |
| $L^*$ / chamber volume                                      | 0.80 m / 12.35 L                 | assumed [1]               |
| Cylinder / convergent length                                | 182 / 136 mm                     | 30° cone, $R_1 = 1.5 r_t$ |
| Injector face to throat / throat to exit                    | 0.318 / 1.968 m                  | 80 % bell                 |
| Bell angles $\theta_N$ / $\theta_E$ ; divergence efficiency | 37° / 8°; 0.9885                 | MOC                       |
| Engine length (gimbal plane to exit, incl. 0.35 m head)     | 2.636 m                          | envelope 2.642 m          |

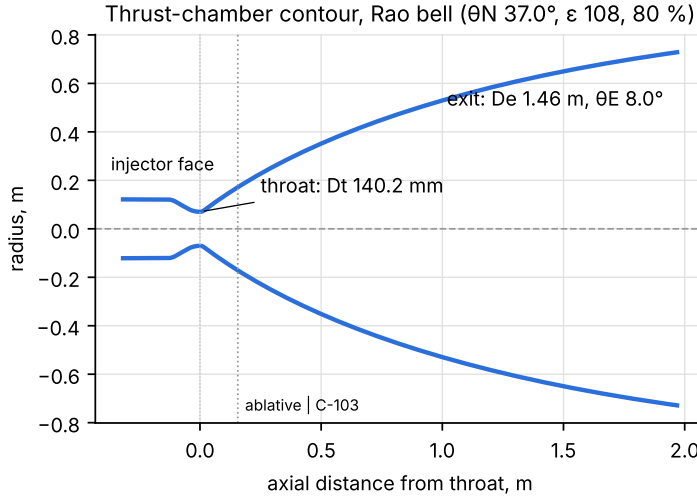


Figure 1: Thrust-chamber contour from the injector face to the nozzle exit.

## 6 Thermal Design

The gas-side heat-transfer coefficient follows Bartz [3; 1, Eq. 4-13]; at the throat  $h_g = 2,286$  W/(m<sup>2</sup>K). The 3,042 K core gas is kept off the wall by running the injector's outer ring fuel-rich: 12.5 % of the flow at O/F 0.60 (1,541 K gas), with the core at O/F 1.92. This is the smallest ring flow that keeps the throat surface below the 1,900 K wear onset of silica phenolic with the heat flux increased by 30 %; it costs 0.78 s of  $I_{sp}$ . The throat heat flux with the ring is 3.1 MW/m<sup>2</sup> into a 700 K wall (Fig. 2).

The chamber and throat are lined with tape-wrapped silica phenolic to area ratio 6. Silica phenolic was chosen over carbon phenolic because it resists oxidation by the H<sub>2</sub>O and CO<sub>2</sub> in NTO/MMH exhaust and insulates better. Beyond area ratio 6 the wall is a radiation-cooled C-103 niobium skirt with an R512E silicide coating. An insulating ring between the liner and the injector limits soak-back after the longest burn. Results are in Table 7 and Figs. 3–4. The liner life is an analytical prediction based on assumed ablator properties and the Bartz estimate. It requires material characterization and hot-fire testing.

Table 7: Thermal design results (bounding case: heat flux  $\times 1.3$ ).

| Item                                      | Value                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------|
| Liner extent                              | injector face to area ratio 6 (0.474 m)                                       |
| Liner thickness / mass                    | 82.6 mm / 66.4 kg                                                             |
| Design firing time                        | 3,305 s ( $1.25 \times 2,644$ s cumulative)                                   |
| Throat surface temperature                | 1,806 K (wear onset 1,900 K); throat recession 0                              |
| Maximum recession                         | 1.38 mm at 146 mm downstream of the throat (area ratio 5.8)                   |
| Char depth at end of life                 | 50.9 mm                                                                       |
| Titanium case temperature at end of life  | 354 K (bond-line limit 420 K)                                                 |
| C-103 extension peak temperature          | 1,389 K nominal, 1,477 K bounding (limit 1,600 K)                             |
| Injector after the longest burn (1,250 s) | 399 K with insulating ring (MMH boils at 442 K at 1 MPa); about 505 K without |

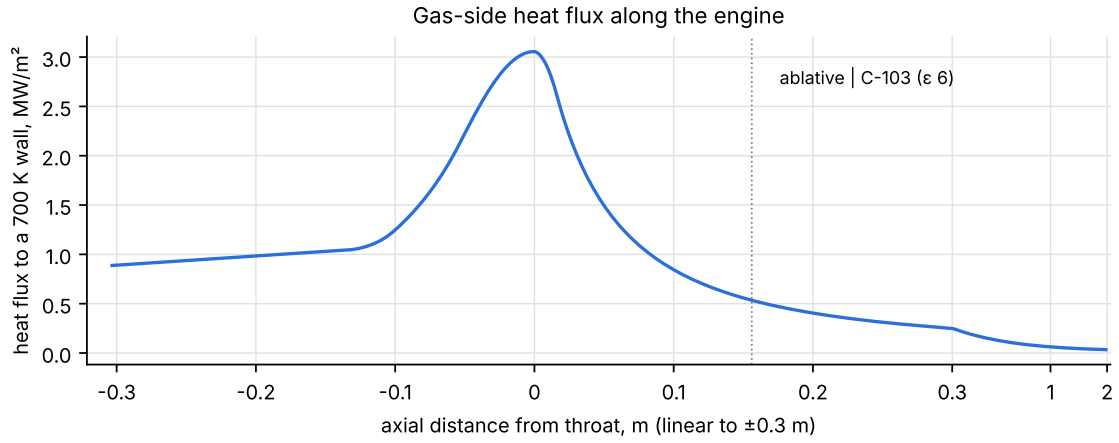


Figure 2: Gas-side heat flux along the engine.

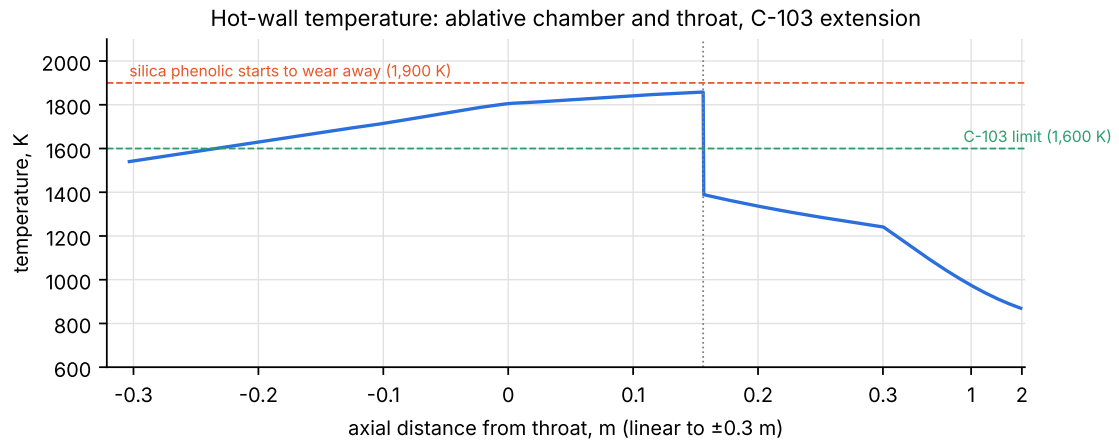


Figure 3: Hot-wall temperature: silica-phenolic liner (left of the dotted line) and C-103 extension.

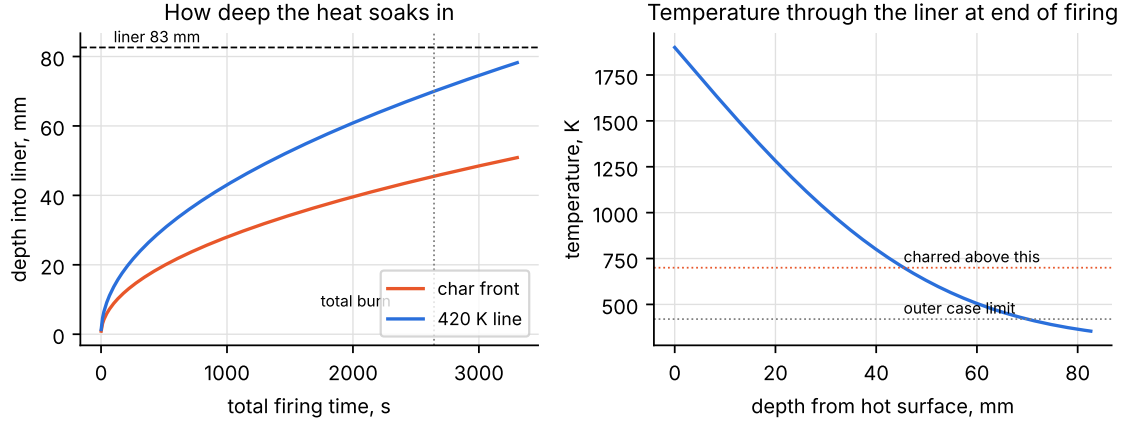


Figure 4: Silica-phenolic liner temperature through the wall at the throat.

**Nozzle extension wall.** The C-103 wall is 0.6 mm thick. Internal pressure is 19.9 kPa at the attach station (Mach 3.0) and 0.44 kPa at the exit, so the hoop stress is at most 5.7 MPa, far below the strength of C-103 at 1,389 K. A radiation-cooled wall runs at radiation equilibrium whatever its thickness, so thickness does not affect temperature. The governing loads are stiffness, vibration, gimbal and handling loads, and the attach flange. These are covered only by a 25 % mass allowance for stiffening rings, flange and coating, so the extension structure is preliminary until checked against the mechanical load cases.

## 7 Injector

Unlike-doublet NTO-MMH elements are used. The NTO orifices take 20 % of the injector-end pressure; the MMH orifices are sized for best mixing by Rupe's criterion  $\rho_o V_o^2 d_o = \rho_f V_f^2 d_f$ , giving 26 %. Both exceed 15 %, which decouples the feed system from the combustion. The element flows sum to the engine totals (Table 8).

Table 8: Injector design and flow balance.

| Parameter                                     | Core          | Outer ring       | Total         |
|-----------------------------------------------|---------------|------------------|---------------|
| Elements                                      | 120           | 48               | 168           |
| NTO / MMH per element, g/s                    | 38.11 / 19.81 | 7.76 / 12.93     | —             |
| NTO / MMH flow, kg/s                          | 4.574 / 2.377 | 0.372 / 0.621    | 4.946 / 2.998 |
| Flow fraction / O/F                           | 87.5 % / 1.92 | 12.5 % / 0.60    | 100 % / 1.65  |
| Orifice diameter NTO / MMH, mm                | 1.753 / 1.338 | 0.791 / 1.081    | —             |
| Jet velocity NTO / MMH, m/s                   | 15.7 / 23.0   | 15.7 / 15.6      | —             |
| $\Delta P$ / injector-end pressure, NTO / MMH | 20 % / 26 %   | shared manifolds | —             |

Table 9: Acoustic modes and cavities.

| Quantity                                            | Value                 |
|-----------------------------------------------------|-----------------------|
| Chamber sound speed, m/s                            | 1,236                 |
| 1T / 1R / 1L frequency, Hz                          | 2,984 / 6,210 / 1,947 |
| Quarter-wave cavities (tuned to 1T), number / depth | 12 / 59.4 mm          |
| Injector mass (304L), kg                            | 10.0                  |

## 8 Feed and Pressurization System

### 8.1 Pressure ladder

Pressures are referenced as follows:

- **Chamber pressure:**  $P_c = 0.862$  MPa, the nozzle-entrance stagnation pressure.
- **Injector-end pressure:** 0.8827 MPa, which is  $P_c$  plus the Rayleigh loss.
- **Injector orifice drops:** referenced to the injector-end pressure.

Each component loss is added from the chamber back to the tank, separately for each propellant (Table 10, Fig. 5). Both tanks run at one regulated ullage pressure, 1.279 MPa; trim orifices absorb the difference between the two sides. The tank maximum expected operating pressure (MEOP) is 1.15 times the ullage pressure, 1.47 MPa, set by the relief-valve crack pressure (1.10 times) plus tolerance.

Table 10: Pressure ladder at the design point, MPa (upstream pressure of each element).

| Station                                                   | NTO         | MMH   |
|-----------------------------------------------------------|-------------|-------|
| Chamber, nozzle-entrance stagnation                       | 0.862       | 0.862 |
| Injector end (chamber side of the injector face)          | 0.883       | 0.883 |
| Injector manifold (orifice $\Delta P$ 177 / 231 kPa)      | 1.059       | 1.114 |
| Engine valve inlet ( $\Delta P$ 40 kPa)                   | 1.099       | 1.154 |
| Trim orifice inlet ( $\Delta P$ 60 / 20 kPa)              | 1.160       | 1.174 |
| Filter inlet ( $\Delta P$ 20 kPa)                         | 1.180       | 1.194 |
| Gimbal flex line inlet ( $\Delta P$ 15 kPa)               | 1.195       | 1.209 |
| Feed line inlet (line $\Delta P$ 64 / 50 kPa)             | 1.259       | 1.259 |
| Latch valve inlet ( $\Delta P$ 15 kPa)                    | 1.274       | 1.274 |
| Tank outlet / PMD ( $\Delta P$ 5 kPa) = tank ullage       | 1.279       | 1.279 |
| Helium regulator outlet                                   | 1.329       |       |
| Tank MEOP / relief crack                                  | 1.47 / 1.41 |       |
| Minimum usable helium bottle pressure (regulator dropout) | 2.33        |       |

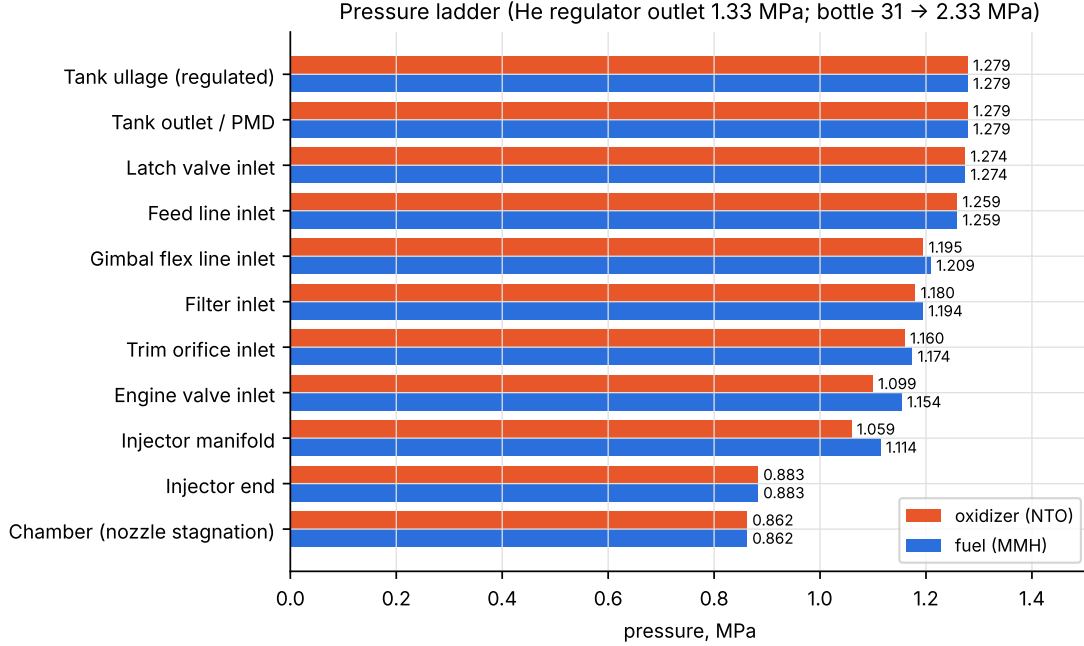


Figure 5: Pressure ladder at the design point.

## 8.2 Feed lines and valve-closure surge

Both feed lines are 1.5 in  $\times$  0.065 in 321 stainless tube, sized for at most 4 m/s (Table 11). On engine shutdown the valves close in 50 ms, longer than the pressure-wave round trip, so the surge is reduced below the instantaneous (Joukowsky) value. The surge is a pressure increase at the closing valve. The tank ullage is held at constant gas pressure, so the wave is relieved at the tank outlet and the tanks do not see it. The components between the tank outlet and the engine valve are therefore rated to a line design pressure of 2.2 MPa, which is the tank MEOP plus the larger surge. At that pressure the tube hoop stress is 23 MPa against 205 MPa yield, a factor of 9.0.

Table 11: Feed lines.

| Parameter                                                    | NTO                | MMH                |
|--------------------------------------------------------------|--------------------|--------------------|
| Routed length (CAD), m                                       | 0.5                | 5.6                |
| Tube OD $\times$ wall, in                                    | 1.5 $\times$ 0.065 | 1.5 $\times$ 0.065 |
| Velocity, m/s                                                | 3.6                | 3.6                |
| Line and fittings $\Delta P$ , kPa                           | 64                 | 50                 |
| Surge increment at 50 ms closure, MPa                        | 0.42               | 0.69               |
| Peak pressure at the engine valve (valve inlet + surge), MPa | 1.52               | 1.85               |
| Line design pressure (MEOP + largest surge), MPa             | 2.2                |                    |

The NTO pressure drop was computed for an assumed 2 m line and is conservative for the 0.5 m routed length.

## 8.3 Propellant tanks

Each tank holds the full propellant load at the specified density plus 5 % ullage (3 % gas space and 2 % liquid expansion from 293 to 305 K) and 0.5 % PMD displacement. Wall thickness uses a burst factor of 2.0 on the 1.47 MPa MEOP. Shell mass is the membrane mass  $\times$  1.30 for welds, bosses, girth ring and mounts (Table 12).

Table 12: Propellant tanks (baseline used in the analysis and the CAD).

| Parameter                                                      | NTO (lower)     | MMH (upper)     |
|----------------------------------------------------------------|-----------------|-----------------|
| Propellant mass, kg                                            | 13,075          | 7,925           |
| Propellant volume at 293 K, m <sup>3</sup>                     | 9.080           | 9.057           |
| Tank volume, m <sup>3</sup>                                    | 9.580           | 9.555           |
| Inner diameter, m                                              | 2.635           | 2.633           |
| Wall (Ti-6Al-4V), mm                                           | 2.16            | 2.16            |
| MEOP, MPa                                                      | 1.47            | 1.47            |
| Free gas volume at 283 / 293 / 305 K, % of tank                | 6.2 / 4.8 / 2.9 | 6.2 / 5.2 / 3.9 |
| Shell (membrane 209.5 / 209.0 kg $\times$ 1.30) / PMD mass, kg | 272 / 15        | 271 / 15        |

## 8.4 Helium pressurization

At end of burn both tanks are full of helium at 1.279 MPa (19.1 m<sup>3</sup>). The required helium is computed for the cold design case, 273 K with adiabatic bottle expansion, using real-gas compressibility in the bottles. A 10 % margin is added for leakage and pre-pressurization (Table 13). The stored 86.5 kg is the initial inventory. At 31 MPa and 273 K the compressibility factor is  $Z = 1.16$ , so 86.5 kg occupies 1.838 m<sup>3</sup>; an ideal-gas estimate would put 100 kg in the same volume. A time-stepped blowdown of the bottles (adiabatic, 273 K start, tanks held at 1.279 MPa) ends at 5.96 MPa, above the 2.33 MPa regulator dropout, with 31.3 kg left in the bottles (Fig. 6).

Table 13: Helium system.

| Item                                                | Value                                                                                |
|-----------------------------------------------------|--------------------------------------------------------------------------------------|
| Required, isothermal / adiabatic bottles, 273 K, kg | 47.2 / 78.6                                                                          |
| Initial inventory (adiabatic 273 K + 10 %), kg      | 86.5                                                                                 |
| Bottles                                             | 4 COPVs, 0.460 m <sup>3</sup> each (1.838 m <sup>3</sup> ), 0.957 m diameter, 31 MPa |
| Bottle mass (all four), kg                          | 380                                                                                  |
| Bottle pressure at end of burn (simulated)          | 5.96 MPa (dropout 2.33 MPa)                                                          |
| Helium delivered / remaining, kg                    | 55.2 / 31.3                                                                          |

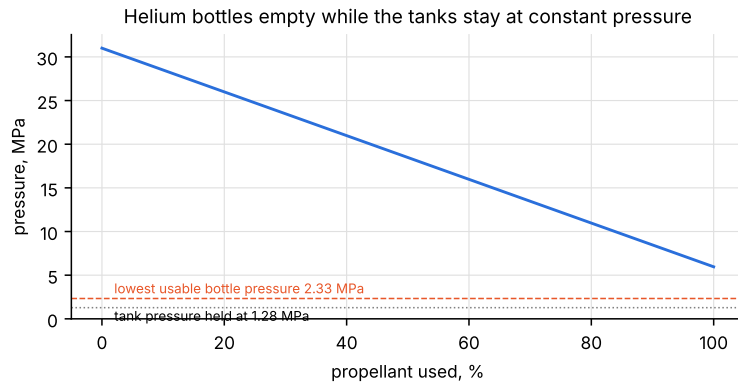


Figure 6: Helium bottle pressure as the propellant is used (design case).

## 9 Materials

Table 14: Materials and margins (allowables assumed, to be verified).

| Part                   | Material                         | Operating condition                         | Allowable                             | Margin                     |
|------------------------|----------------------------------|---------------------------------------------|---------------------------------------|----------------------------|
| Propellant tanks       | Ti-6Al-4V                        | 448 MPa membrane stress at MEOP             | 895 MPa ultimate                      | factor 2.00 (2.0 required) |
| Chamber/throat liner   | silica phenolic                  | 1,806 K surface                             | 1,900 K wear onset                    | 94 K                       |
| Liner back face / case | Ti-6Al-4V                        | 354 K; 108 MPa                              | 420 K bond line; 828 MPa yield        | 66 K; factor 7.7           |
| Nozzle extension       | C-103 + R512E                    | 1,389 K                                     | 1,600 K                               | 211 K                      |
| Injector               | 304L                             | 399 K after shutdown                        | 442 K (MMH boiling at 1 MPa)          | 43 K                       |
| Feed lines             | 321 stainless                    | 2.2 MPa design pressure; 23 MPa hoop stress | 205 MPa yield                         | factor 9.0                 |
| Helium bottles         | COPV (Ti liner, carbon overwrap) | 31 MPa                                      | burst $\geq 1.5 \times$ MEOP (vendor) | vendor item                |
| Seals, valve seats     | PTFE / PCTFE                     | 283–305 K, NTO and MMH                      | 73–533 K (PTFE)                       | large                      |

NTO must be procured to MIL-PRF-26539 with controlled nitric-oxide content to avoid stress-corrosion cracking of the titanium tanks.

## 10 Mass Budget

The line mass uses the routed CAD lengths (Sec. 11), 2.5 kg more than the original allowance.

Table 15: Propulsion-system mass budget.

| Engine                                                                     | kg           | Feed and pressurization        | kg             |
|----------------------------------------------------------------------------|--------------|--------------------------------|----------------|
| Injector                                                                   | 10.0         | NTO tank                       | 271.9          |
| Valves                                                                     | 14.0         | MMH tank                       | 271.2          |
| Liner                                                                      | 66.4         | PMDs                           | 30.0           |
| Case                                                                       | 10.0         | COPVs                          | 379.9          |
| Extension (32.5 bare $\times$ 1.25)                                        | 40.7         | Helium                         | 86.5           |
| Gimbal actuators                                                           | 20.0         | Components (Table 16)          | 40.0           |
| Miscellaneous                                                              | 10.0         | Lines (CAD)                    | 15.4           |
|                                                                            |              | Supports                       | 109.3          |
| <b>Engine</b>                                                              | <b>171.1</b> | <b>Feed and pressurization</b> | <b>1,204.3</b> |
| <b>Propulsion system: 1,375 kg, 21.2 % of the 6,500 kg structural mass</b> |              |                                |                |

Table 16: Components allowance (allocated, to be replaced by vendor data).

| Item                                                    | kg        | In CAD          |
|---------------------------------------------------------|-----------|-----------------|
| Helium regulators (2, in series)                        | 6         | regulator panel |
| Helium check valves (2)                                 | 4         | regulator panel |
| Helium latch / isolation valves (2), fill valve, filter | 7         | budget only     |
| Relief valves (2)                                       | 3         | budget only     |
| Propellant latch valves (2)                             | 10        | budget only     |
| Propellant filters (2)                                  | 3         | budget only     |
| Propellant fill, drain and vent valves (4)              | 4         | budget only     |
| Transducers and fittings                                | 3         | budget only     |
| <b>Total</b>                                            | <b>40</b> | panel = 10 kg   |

## 11 CAD Model

The propulsion bay is modelled in Onshape as one parametric FeatureScript feature, generated directly from the analysis outputs (engine contour and layout). It contains:

- the engine: liner, case and C-103 extension as smooth bodies of revolution, the injector plate with 12 acoustic cavities, and the head, valve and gimbal stack;
- both tanks and the four COPVs;
- the feed and helium lines, and the regulator panel;
- reference shells for the SM, the capsule and the launch escape system (LES), traced from the PDR drawing [6].

A *Cut view* option removes the  $-Y$  half of every body (Figs. 9–11). Units are millimetres; the origin is at the SM base on the axis, with  $+Z$  up. Lumped items (head stack, COPVs, regulator panel) carry their budget masses. The acoustic cavities are modelled as  $\varnothing 16$  mm bores on a 100 mm pitch radius.

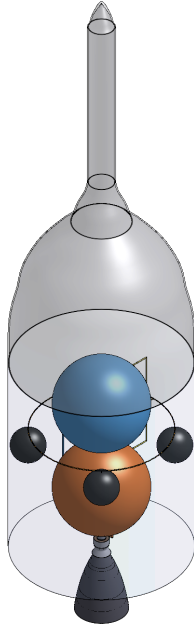


Figure 7: Full model, isometric.

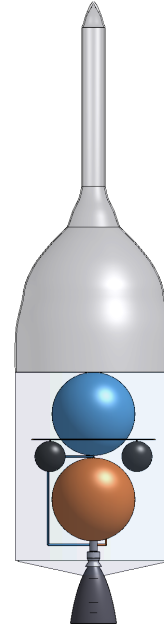


Figure 8: Full model, front.

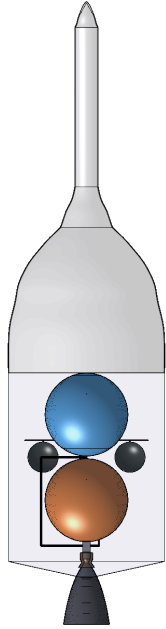


Figure 9: Cut view, front.

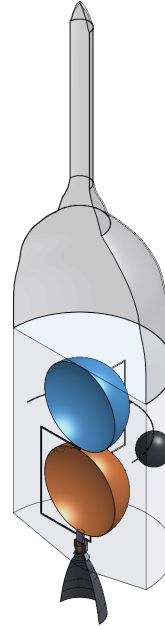


Figure 10: Cut view, isometric.

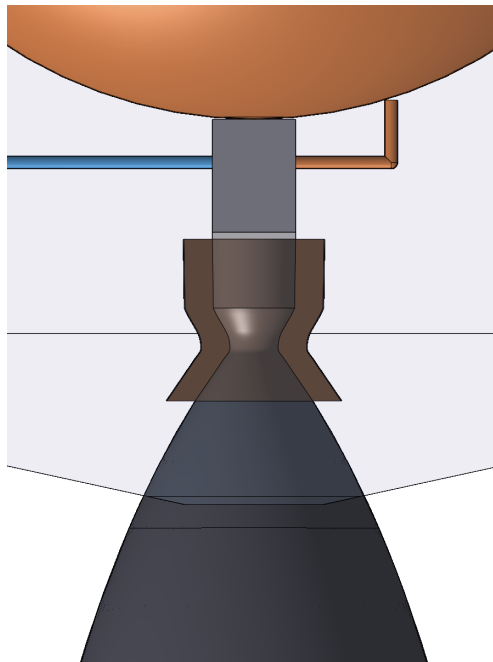


Figure 11: Cut view of the engine with the NTO (right) and MMH (left) feed lines entering the valve stack.

Table 17: Layout and clearances (from the CAD model).

| Item                                                     | Value                                                                                                |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Gimbal plane / injector face / throat / nozzle exit, $z$ | 626 / 276 / $-42$ / $-2,010$ mm                                                                      |
| NTO tank centre; outer diameter                          | $z = 1,946$ mm; 2.639 m                                                                              |
| MMH tank centre; outer diameter                          | $z = 4,674$ mm; 2.637 m                                                                              |
| Gap between tanks (MMH line passes through it)           | 90 mm (26 mm each side of the line)                                                                  |
| Tank stack, gimbal plane to MMH top                      | 5.366 m; top at $z = 5,992$ mm, 7.6 mm below the SM top                                              |
| COPVs                                                    | 4 at $45/135/225/315^\circ$ , radius 1,961 mm, $z = 3,315$ mm                                        |
| COPV to SM wall / to tanks                               | 50 mm / 589 mm (radial space beside the tanks: 1.17 m)                                               |
| Feed line routes                                         | NTO: lower dome to $+X$ side of the valve stack; MMH: bottom pole, tank gap, down at $x = -1,450$ mm |
| Helium lines                                             | 0.375 in HP ring ( $z = 3,854$ mm) to the regulator; 0.75 in LP lines to each upper dome             |
| Interference (136 body pairs)                            | none; connected parts meet within 0–0.12 mm                                                          |
| Outside the SM envelope                                  | C-103 extension only (exit 2.01 m below the SM base, by design)                                      |

The CAD masses agree with the budget once undrawn items are accounted for:

- **Liner:** 66.1 kg in the CAD against 66.4 kg in the budget.
- **Bare tank membranes:** 209.5 and 209.0 kg; times 1.30 these match the 272 and 271 kg budget.
- **Bare extension wall:** 33.2 kg against 32.5 kg calculated.
- **Case:** 6.5 kg; the budget adds 3 kg of flanges.
- **Injector plate:** 6.9 kg; the budget adds a 2.6 kg manifold.
- **Lines:** 15.4 kg, now carried in the budget.

## 12 Requirements Compliance

Table 18: Design against the Crew Capsule requirements.

| Requirement                                      | Required                       | Design (status)                                                   |
|--------------------------------------------------|--------------------------------|-------------------------------------------------------------------|
| Thrust, engines                                  | 25 kN, 1                       | 25 kN vacuum, 1 (met)                                             |
| Specific impulse                                 | $\geq 280$ s                   | 320.8 s nominal, 303.5 s pessimistic (met, predicted)             |
| $\Delta v$ with $M_0$ 38,000 kg, $M_b$ 17,000 kg | 2,209 m/s                      | 2,531 m/s nominal, 2,394 m/s pessimistic (met, predicted)         |
| Propellant load                                  | 21,000 kg at O/F 1.65          | fits with 2.9 % free gas volume at the 305 K hot case (met)       |
| $T/W$ at the Moon                                | 0.405 / 0.905                  | unchanged (met)                                                   |
| Firing life                                      | 2,644 s cumulative, 11 starts  | liner 3,305 s predicted; 22 design starts (preliminary)           |
| SM envelope                                      | 6 m section; 2.01 m below base | stack 5.366 m (7.6 mm clearance); engine 2.636 m of 2.642 m (met) |
| Propulsion mass                                  | within 6,500 kg $M_s$          | 1,375 kg, 21.2 % (met)                                            |

## 13 Conclusions and Open Items

The 25 kN pressure-fed NTO/MMH engine is predicted to deliver 320.8 s vacuum  $I_{sp}$  (303.5 s pessimistic) at 0.862 MPa chamber pressure, O/F 1.65 and area ratio 108, with an 80 % Rao bell of 1.457 m exit diameter. A silica-phenolic chamber protected by a fuel-rich outer ring, followed by a C-103 extension, is predicted to survive the 2,644 s cumulative firing with margin. The tanks, the four helium bottles and the engine fit the PDR SM envelope, and the tanks hold the full propellant load at the hot case. The propulsion system mass is 1,375 kg.

These are analytical results. The following items remain open:

1. Heat flux, fuel-rich-ring effectiveness and ablator properties are uncalibrated. The liner life must be confirmed by material characterization and hot-fire tests, including restart thermal cycling.
2. The nozzle-extension structure (wall, attach flange and stiffening rings) must be analysed for vibration, gimbal and handling loads.
3. The feed-line transient must be confirmed with a system water-hammer model and component ratings of at least 2.2 MPa.
4. Feed lines need flexible sections across the gimbal; the components allowance needs vendor masses.
5. The tank stack leaves 7.6 mm below the SM top; the capsule interface must be checked.
6. NTO must be procured to MIL-PRF-26539 with controlled nitric-oxide content.

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